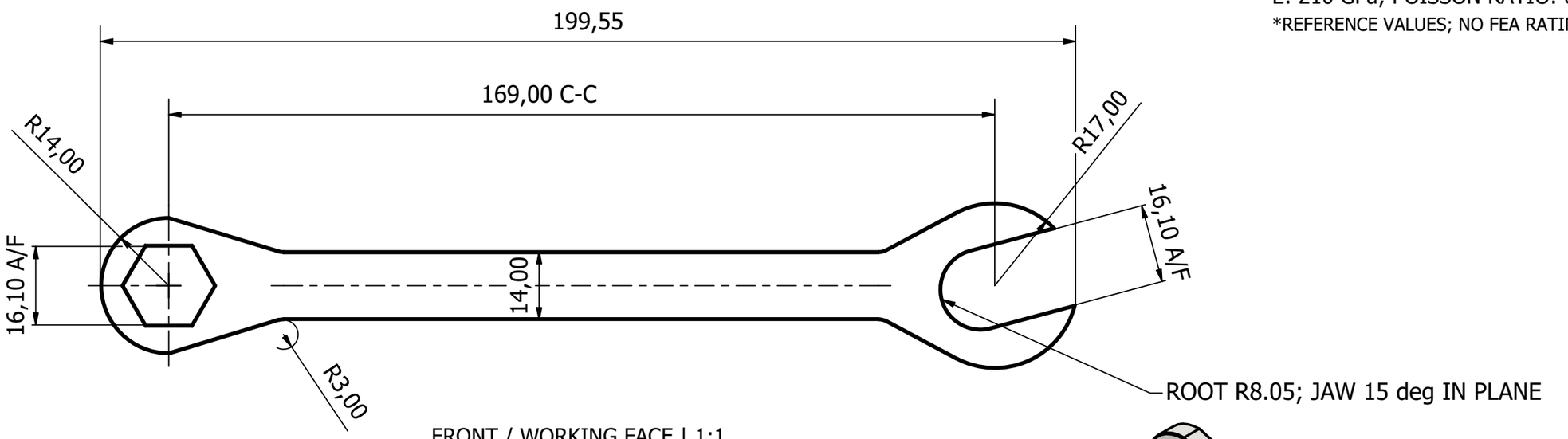
16 mm COMBINATION WRENCH

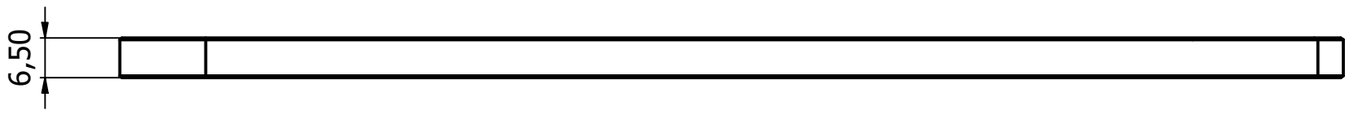
SIX-POINT RING / 15 deg OPEN JAW / FLAT PATTERN

PART: WR-016-200 | REV A

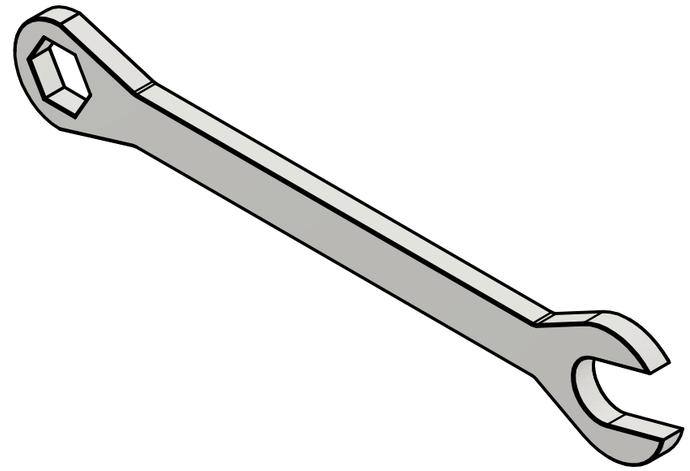
MATERIAL: AISI 6150 Cr-V STEEL
CALCULATED MASS: 148.7 g
DENSITY: 7850 kg/m^3 (library)
E: 210 GPa; POISSON RATIO: 0.30*
*REFERENCE VALUES; NO FEA RATING



FRONT / WORKING FACE | 1:1



BOTTOM / EDGE VIEW | 1:1



ISOMETRIC | 3:4

MANUFACTURING / DESIGN REQUIREMENTS

1. FORGE AISI 6150; QUENCH AND TEMPER TO 44-48 HRC.
2. SATIN NICKEL/CHROME PLATE; FINISH OPENINGS AFTER PLATING.
3. BOTH OPENINGS: 16.10 +0.05/-0.00 A/F, THROUGH.
4. EDGE BREAK: 0.3 x 45 deg BOTH FACES; REMOVE ALL BURRS.
5. UNLESS NOTED: LINEAR +/-0.2; ANGLES +/-0.5 deg.
6. RING OFFSET: 0 deg. OPEN JAW: 15 deg IN PLANE.
7. NOMINAL CAD DESIGN; STRENGTH / TORQUE NOT VALIDATED.

16 mm COMBINATION WRENCH WR-016-200 | AISI 6150 | SATIN CHROME UNITS: mm | THIRD ANGLE | DO NOT SCALE	2026-11-07 | REV A	A3 | SHEET 1 OF 2
	SCALE: AS SHOWN	DESIGN MODEL

PROFILE & PARAMETER SCHEDULE

WR-016-200 | REV A | UNITS mm | SHEET 2 OF 2

NOMINAL BLANK / MACHINING DEFINITION

DATUM: RING-HEAD OUTER LEFT EXTREME X=0;
BODY CENTRELINE Y=0.
Z=0 IS LOWER FACE. BODY EXTRUDES TO Z=6.50.
BLANK ENVELOPE BEFORE JAW CUT: 200 x 34 x 6.5.
RING CENTRE: (14,0), OUTER R14.
OPEN-HEAD CENTRE: (183,0), OUTER R17.
JOIN THE FOLLOWING POINTS IN ORDER (UPPER HALF):
(14,14) -> (37,7) -> (160,7) -> (175,15).
MIRROR THESE LINES ABOUT Y=0 FOR THE LOWER HALF.
CLOSE LEFT END WITH R14 ARC THROUGH (0,0).
CLOSE RIGHT END WITH R17 ARC THROUGH (200,0).
ADD R3 BLENDS AT (37,+/-7) AND (160,+/-7).

OPENINGS / EDGE FINISH

RING: REGULAR SIX-POINT HEXAGON CENTRED AT (14,0).
TWO FLATS PARALLEL TO X; ACROSS FLATS = 16.10.
OPEN JAW: LOCAL ORIGIN (183,0), LOCAL X ROTATED +15 deg.
PARALLEL CONTACT FLATS AT LOCAL Y=+/-8.05.
FLATS START AT LOCAL X=-3 AND CONTINUE OUTSIDE THE HEAD.
JOIN FLAT ROOTS WITH LEFT SEMICIRCLE R8.05, CENTRE (-3,0).
CUT BOTH OPENINGS THROUGH THE BODY.
CHAMFER UPPER AND LOWER PERIMETERS 0.3 x 45 deg.
ALL COORDINATES NOMINAL BEFORE EDGE BREAK; FINAL OPENING LIMITS APPLY TO CONTACT LANDS AFTER PLATING.

MATERIAL / PROCESS REQUIREMENTS

MATERIAL: AISI 6150 CHROMIUM-VANADIUM STEEL.
INVENTOR MATERIAL ASSET: STEEL AISI 6150.
DENSITY: 7850 kg/m^3.
ELASTIC MODULUS: 210 GPa (REFERENCE).
POISSON RATIO: 0.30 (REFERENCE).
QUENCH / TEMPER: TARGET 44-48 HRC.
SATIN NICKEL / CHROME FINISH.
HARDNESS AND FINISH ARE DESIGN REQUIREMENTS.
LIBRARY ASSET DOES NOT CERTIFY HEAT TREATMENT.

MATCHING EXISTING BOLT

FILE: bolt.ipt | ISO 4017 NOMINAL FORM.
THREAD: M10 x 1.5 - 6g, RIGHT HAND.
UNDER-HEAD LENGTH: 30.0; HEAD HEIGHT: 6.4.
HEAD ACROSS FLATS: 16.0.
MATERIAL: CARBON STEEL, QUENCHED AND TEMPERED.
STRENGTH REQUIREMENT: ISO 898-1 CLASS 8.8.
WRENCH NOMINAL A/F CLEARANCE: 0.10.
BOLT THREAD IS AN INVENTOR COSMETIC THREAD.
SEE bolt_drawing.idw / .pdf FOR BOLT DIMENSIONS.

DESIGN BASIS / RELEASE STATUS

CONVENTIONAL COMBINATION-WRENCH FORM WITH A FLAT SIX-POINT RING.
THIS MODEL IS NOT A COPY OF A CERTIFIED DIN / ISO TOOL. BEFORE MANUFACTURE OR SERVICE, VALIDATE TORQUE CAPACITY, FIT, HEAT TREATMENT AND COATING PROCESS. DO NOT SCALE THE DRAWING.